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3D medical image segmentation via sequential 2D slice processing

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ABSTRACT

Three-dimensional (3D) medical image segmentation is vital in clinical applications, but faces challenges like computational complexity and spatial context loss in 2D processing. To address these issues, we propose 3DAS2D, which processes 3D images as 2D slices while retaining volumetric data for better accuracy. Our approach integrates modules like Neighbour Attention, Memory Attention, Memory Bank, and Mask Encoder to capture inter-slice dependencies and historical context within a 2D prediction framework. Evaluated on four datasets covering prostate, cardiac, and lung tumour tasks, 3DAS2D performed comparably to 3D methods. This work offers an efficient, adaptable solution bridging 2D efficiency and 3D accuracy, with significant clinical potential.

1. Introduction

Medical image segmentation, which involves partitioning critical regions of interest, is essential for diagnosis, treatment planning, and image-guided surgery applications [1]. Although the research community has significantly enhanced the performance of deep learning models in medical image segmentation [2,3], substantial challenges persist. A primary concern is the generalisation of models across 2D and 3D imaging modalities. Many deep learning architectures are tailored for 2D images [4,5], posing a significant challenge when adapting them to 3D medical images. This adaptation is crucial for multimodal learning [6] and transfer learning [7]. Moreover, 3D deep learning models typically demand greater computational resources than their 2D counterparts.

The common approaches to 3D medical image segmentation can be broadly categorised into two types: processing as volumetric 3D images and processing as 2D slice images. In the 3D processing approach, deep learning models [3,8] are specifically designed to accept volumetric 3D images as input, enabling them to preserve the spatial and contextual information inherent in the data. Conversely, the 2D processing [9,10] approach involves slicing 3D images into multiple 2D slices, which are then processed using 2D deep learning models. While this method substantially reduces computational requirements, the spatial information of the 3D images is often compromised, as inter-slice relationships are not fully captured.

To address these challenges, we introduce 3DAS2D (3D as 2D). This model utilises a 2D deep learning architecture to process 3D medical images as a sequence of 2D slices while preserving the spatial information of the 3D volume, thereby enhancing segmentation performance. We incorporate modules such as Neighbour Attention, Memory Attention, Memory Bank, and Mask Encoder to integrate the 3D spatial context into 2D segmentation, effectively bridging the gap between computational efficiency and volumetric fidelity.

We evaluated 3DAS2D against leading 2D, 2.5D, and 3D segmentation methods across four datasets. The results demonstrate that our proposed method achieves performance comparable to 3D methods that fully incorporate spatial information, while maintaining lower computational demands. In summary, this paper presents two key contributions: First, we introduce 3DAS2D, a 2D segmentation framework that processes 3D medical images as 2D sequences by integrating 3D spatial information to enhance segmentation accuracy. Second, we demonstrate that 3DAS2D achieves performance comparable to 3D methods, suggesting its potential for adapting existing off-the-shelf 2D large vision models to 3D medical image segmentation tasks without requiring any architectural modifications or retraining.

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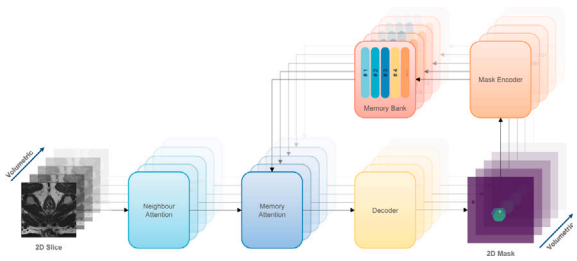

Fig. 1. Overall architecture of 3DAS2D.

2. Related works

2.1. 2.5D segmentation

The 2.5D segmentation approach seeks to bridge the gap between 2D and 3D segmentation by utilising the computational efficiency of 2D models while incorporating inter-slice information to capture spatial dependencies more effectively. This is typically achieved by concatenating several consecutive image slices into a single 2D image with multiple channels. A 2D segmentation model then processes this multi-channel image to delineate the region of interest in the central slice [11,12]. While traditional 2.5D methods suffer from misalignment shortcomings, CSA-Net [13] introduces a Cross-Slice Attention module [14] to address this issue. This module effectively captures 3D spatial information by learning long-range dependencies between the central slice and its neighbouring slices. Although this enables the model to discern spatial relationships between the central slice and adjacent slices, it relies solely on the spatial information embedded within the medical image, overlooking the spatial information inherent in the segmentation mask.

2.2. MedSAM2 and SAM2

MedSAM2 [15] is an advanced segmentation model designed for both 2D and 3D medical image segmentation tasks. Building upon the SAM2 [16] framework, MedSAM2 treats medical images as video sequences, enabling the adaptation of video object tracking techniques to enhance segmentation performance. Both MedSAM2 and SAM2 are designed for promptable segmentation, where users delineate the region of interest by providing prompts, such as bounding boxes or points, and the model subsequently segments the designated area.

3. Method

Fig. 1 illustrates the overall architecture of the proposed model. Within this pipeline, 3D images are processed slice by slice. Each 2D slice is fed into the model to produce its corresponding 2D segmentation mask. These 2D masks are subsequently concatenated to reconstruct a complete 3D segmentation mask.

3.1. Neighbour attention

We introduce the Neighbour Attention module, designed to fuse spatial information from neighbouring slices with the current slice. This module leverages both the immediately preceding and subsequent slices, enabling bidirectional integration of local spatial cues. A total number of t neighbouring slices are extracted relative to the nth current slice. This results in three types of inputs to the model: the current slice $X_n \in \mathbb{R}^{H\times W\times C}$; the preceding slices $\{X_{n-t}, X_{n-t+1}, \dots, X_{n-1}\} \in \mathbb{R}^{H\times W\times C}$; and subsequent slices $\{X_{n+1}, X_{n+2}, \dots, X_{n+t}\} \in \mathbb{R}^{H\times WXC}$. These inputs are encoded by a DINOv2 [17] features extractor $\mathcal{E}_{DINO}$, resulting in feature embeddings $\{F_{n-t}, \dots, F_{n+t}\} \in \mathbb{R}^{S\times D}$, where S represents sequence length including a $[class]$ token and D is the embedding dimension. The $[class]$ token is discarded, retaining only the spatial feature patches for subsequent attention operations. This process can be formulated as

$$F_i = \mathcal{E}_{\text{DINO}}(X_i) \quad \text{for} \quad i \in \{n-t, \dots, n+t\} \tag{1}$$

DINOv2 was employed as the feature extractor due to its capacity to acquire robust and highly generalisable visual features via self-supervised learning, thereby obviating the need for task-specific labelled data. This approach enables our framework to leverage powerful pre-trained representations, furnishing an effective and efficient foundation for the analysis of complex anatomical structures within medical imagery.

After obtaining the image feature embeddings, we implement two distinct attention mechanisms — self-attention (SA) and cross-attention (CA) — both realised through the multi-head attention (MHA) framework proposed by [14]. Self-attention operates exclusively within the current slice, capturing intra-slice spatial relationships to produce an in-slice embedding, denoted as $\mathbf{E}_n$.

In contrast, cross-attention is divided into two variants: preceding CA and subsequent CA. Preceding CA performs cross-attention between the features of the preceding slices $F_{n-t}, \dots, F_{n-1}$ and the current slice F_n, producing the preceding slice embeddings $\{\mathbf{E}_{n-t}, \dots, \mathbf{E}_{n-1}\}$. Conversely, subsequent CA computes cross-attention between the features of the subsequent slices $F_{n+1}, \dots, F_{n+t}$ and the current slice F_n, producing the subsequent slice embeddings $\{\mathbf{E}_{n+1}, \dots, \mathbf{E}_{n+1}\}$. The number of SA operations is fixed at one, applied solely to the current slice. In contrast, the number of CA operations is proportional to the number of neighbouring slices t. All CA and SA operations are independent and do not share weights.

Following the application of SA and CA within the Neighbour Attention module, the resulting feature embeddings are concatenated and subsequently downsampled to produce an image embedding, $\mathbf{Z} \in \mathbb{R}^{S\times D}$. This process is achieved through two layers of 1×1 convolution, with a ReLU activation function applied between the layers.

3.2. Memory attention

We argue that volumetric information is not only present in the input 3D medical imaging data but also manifests in the segmentation mask. Therefore, a Memory Attention module is designed to condition the features of the current slice on preceding predictions, thereby capturing temporal dependencies that enhance volumetric consistency. As illustrated in Fig. 2, this module comprises L stacked image-to-memory (cross-attention) blocks, which integrate the image embedding $\mathbf{Z}$ derived from the Neighbour Attention module with the mask embedding retrieved from the Memory Bank. Subsequently, a stack of twelve self-attention layers refines these integrated features, enhancing intra-slice coherence and preparing them for downstream decoding into segmentation masks.

3.3. Memory bank

A Memory Bank is utilised to store mask embeddings for the Memory Attention module. These embeddings encapsulate information from preceding predictions. The bank operates on a first-in-first-out (FIFO) basis, retaining the mask embeddings from the L most recent slices. However, a challenge arises during the initial processing stages, specifically for the first few slices of a volumetric scan. In these early stages, the memory bank is either empty or contains insufficient information to execute memory attention effectively, which can compromise the model's performance. To mitigate this issue, the memory bank is initialised with zero-maps (zero-value embeddings), which are subsequently replaced according to the FIFO sequence as new slice embeddings are generated. The efficacy of this padding strategy is evaluated in an ablation study presented in Section 6.2.

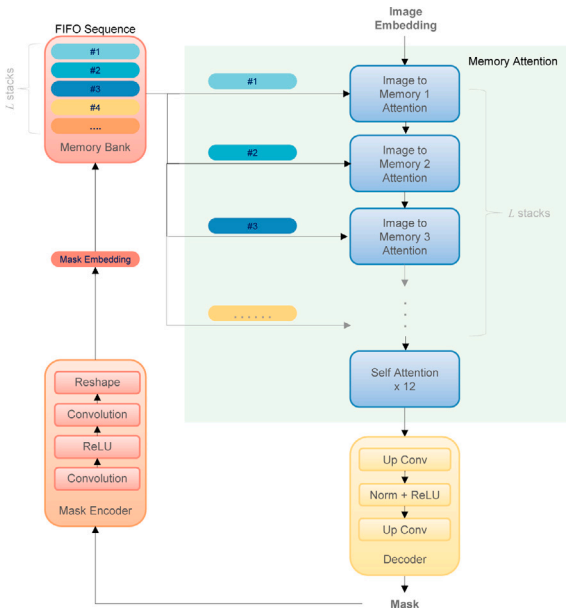

Fig. 2. Memory attention module, decoder, mask encoder and memory bank.

3.4. Decoder

We employ a decoder comprising five convolutional blocks to generate 2D segmentation masks from the processed embeddings. Each block consists of a transposed convolution followed by a regular convolution, with a ReLU activation function applied between them to introduce non-linearity. Before being fed into these convolution blocks, the embedding, represented initially as $\mathbf{Z}'' \in \mathbb{R}^{S \times D}$ is reshaped into a spatial format, $\mathbf{Z}'' \in \mathbb{R}^{\frac{D}{2} \times \frac{H}{16} \times \frac{W}{16}}$, aligning with a reduced-resolution grid.

3.5. Mask encoder

The Mask Encoder generates a memory embedding by downsampling the predicted segmentation mask through a convolutional module comprising two convolutional layers (kernel size 7×7) followed by a reshape operation. The reshape operation reshapes the mask embedding from $\mathbf{E}_{\text{mask}} \in \mathbb{R}^{D \times H \times W}$ to $\mathbf{E}_{\text{mask}} \in \mathbb{R}^{(H \times W) \times D}$. This process encodes the mask into a shape identical to the image embedding, facilitating seamless integration with the Memory Attention module.

4. Experiment

4.1. Dataset

For the purpose of evaluation, we utilised four datasets: ProstateX [18], MSD-Prostate [19], ACDC [20], and MSD-Lung [19].

ProstateX: ProstateX dataset comprises 98 T2-weighted prostate MRI images. Each MRI image features an in-plane resolution of approximately 0.5 mm and a slice thickness of 3.6 mm. The segmentation masks encompass four anatomical classes: the peripheral zone, transition zone, fibromuscular stroma, and distal prostatic urethra. We randomly divided the dataset into 68 training images and 30 testing images for evaluation purposes.

MSD-Prostate: The MSD-Prostate dataset comprises 32 MRI images, including both T2-weighted and apparent diffusion coefficient (ADC) modalities. The segmentation masks delineate two regions: the central gland and the peripheral zone. We only utilised the T2-weighted images for this study, as this modality was deemed sufficient. The dataset was randomly divided into 27 training images and 5 testing images.

ACDC: The ACDC dataset comprises 100 cases, with annotations for the left ventricle (LV), right ventricle (RV), and myocardium (Myo) in the diastolic (ED) and systolic (ES) frames of cine-MRI images of the heart. For our evaluation, we randomly selected 85 cases for training, allocating the remaining 15 for testing.

MSD-Lung: The MSD-Lung dataset comprises 64 CT images from patients with non-small cell lung cancer. For our implementation, we cropped the first 30 slices from each CT image because they lack lung structures. This focused the analysis on relevant anatomical regions. The dataset was then randomly divided into 54 training images and 10 validation images.

4.2. Implementation details

The 3D images are processed as sequences of 2D slices in the axial plane, consistent with the sequential segmentation approach of our 3DAS2D framework. For input to the pre-trained ViT-B/14 DINOv2 feature extractor [17], all images from the datasets are resized to a uniform dimension of 224×224 pixels. Where applicable, their single-channel data are replicated across three channels, yielding an input dimension of $224 \times 224 \times 3$. The parameters of this feature extractor are frozen during training to preserve its pre-trained representations. Additionally, the images are normalised. We applied image augmentation techniques to improve robustness, including horizontal and vertical flips and random 90-degree rotations.

The training and evaluation of our proposed model were conducted on an Nvidia RTX 3090 GPU using the PyTorch and MONAI frameworks. We employed a loss function combining Dice loss and focal loss with equal weight for training. The Adam optimiser was utilised, incorporating a learning rate warmup phase and a cosine annealing scheduler for decay. Across all datasets, a batch size of 6 and 300 epochs were consistently employed. The number of neighbouring slices t was set to 3, and the size of the memory bank L was set to 5.

5. Results

Each dataset in Section 4.1 was trained and assessed independently to ensure a robust performance analysis across diverse medical imaging scenarios. For comparison, we selected UNet [10] and TransUNet [9] as benchmarks for 2D and 3D segmentation, CSA-Net [13] for its hybrid 2.5D approach, and MedSAM2 [15] as a representative of promptable segmentation techniques applied to 3D images processed as video sequences.

5.1. Quantitative results

5.1.1. Prostatex and MSD-Prostate

Tables 1 and 2 present the performance of our proposed model on the ProstateX and MSD-Prostate datasets, benchmarked against state-of-the-art methods. Across both datasets, the results reveal that MedSAM2 achieves the highest DSC across all classes, with average scores of 0.732 and 0.8769, respectively, closely followed by 3DAS2D with DSCs of 0.7266 and 0.8687. A detailed analysis of per-class DSC scores shows that MedSAM2 excels in segmenting the transitional zone and anterior fibromuscular stroma, whereas 3DAS2D demonstrates superior performance on the peripheral zone and distal prostatic urethra. The superior performance of Medsam2 over our method may be attributed to its promptable segmentation approach, which explicitly cues about the region of interest, potentially enhancing its segmentation accuracy. Nevertheless, the results indicate that our model outperforms established 2D, 2.5D, and 3D segmentation methods, showcasing superior accuracy and efficiency across diverse datasets and tasks, as evidenced by its competitive DSC scores. This suggests that our model successfully captures the spatial information of the 3D volume.

Table 1
Dice coefficient and Hausdorff distances (mm) for ProstateX; Avg denotes average; TZ denotes transition Zone; PZ denotes Peripheral Zone; UT denotes Urethra; AT denotes Anterior.

Methods	DSC					HD95				
	Avg	TZ	PZ	UT	AT	Avg	TZ	PZ	UT	AT
UNET (2D)	0.579	0.809	0.651	0.555	0.300	4.04	3.14	4.76	2.73	5.51
UNET (3D)	0.601	0.823	0.676	0.587	0.316	3.80	2.93	4.23	2.69	5.38
TransUNet (2D)	0.631	0.848	0.704	0.593	0.377	3.21	2.29	3.69	2.25	4.63
TransUNet (3D)	0.647	*0.855*	0.713	0.615	0.406	**2.29**	**2.21**	3.57	2.08	4.02
CSA-Net (2.5D)	0.659	0.851	0.720	*0.653*	0.413	2.70	*2.26*	3.43	1.43	3.71
MedSAM2	**0.732**	**0.881**	*0.812*	0.621	**0.614**	2.58	4.12	*2.48*	*1.06*	*2.66*
Ours	*0.727*	0.611	**0.869**	**0.878**	*0.548*	*2.46*	5.20	**1.41**	**1.00**	**2.24**

Table 2
Dice coefficient and Hausdorff distances (mm) for MSD-Prostate; Avg denotes average; TZ denotes Transition Zone; PZ denotes Peripheral Zone.

Method	DSC			HD95		
	Avg	TZ	PZ	Avg	TZ	PZ
UNET (2D)	0.798	0.697	0.900	2.77	1.41	4.12
UNET (3D)	0.809	0.695	0.922	4.50	2.00	7.00
TransUNet (2D)	0.819	0.723	0.914	2.12	1.41	2.83
TransUNet (3D)	0.832	0.737	**0.928**	3.87	1.73	6.00
CSA-Net (2.5D)	0.829	0.759	0.899	2.00	*1.00*	3.00
MedSAM2	**0.877**	**0.837**	*0.917*	*1.98*	1.73	*2.24*
Ours	*0.869*	*0.827*	0.910	**1.62**	**1.00**	**2.24**

5.1.2. ACDC

Table 3 presents the performance of our proposed model compared with other state-of-the-art methods on the ACDC dataset. Our model achieves the highest DSC with a score of 0.9066, surpassing MedSAM2, which records a DSC of 0.8944.

5.1.3. MSD-Lung

Table 4 demonstrates that our proposed 3DAS2D model and MedSAM2 achieve comparable overall DSC, with scores of 0.8321 and 0.8396, respectively. However, in terms of the HD95, our model records a higher value than both CSA-Net and MedSAM2, suggesting that its boundary accuracy is less precise.

5.2. Qualitative results

Fig. 3 illustrates a qualitative comparison of our proposed 3DAS2D model against state-of-the-art methods for the selected datasets.

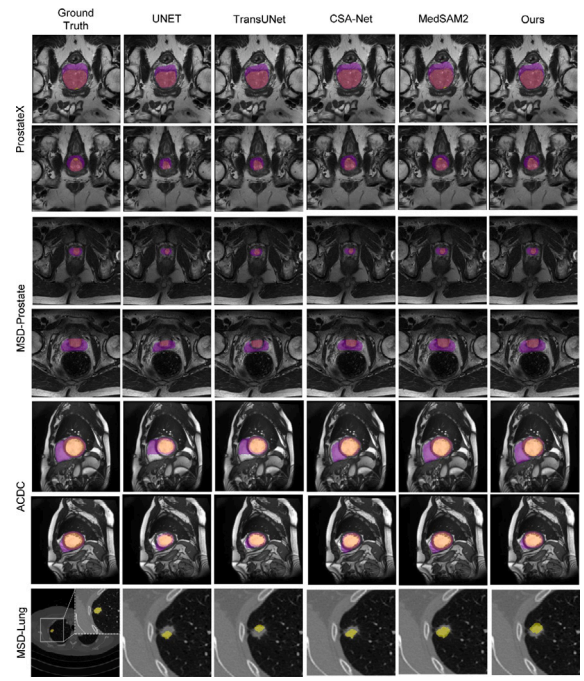

Fig. 3. Qualitative comparison on chosen dataset; Results shown for MSD-Lung are zoomed-in section, full images were used in the training and testing phase.

6. Ablation study

6.1. Importance of model's modules

To assess the contribution of each module to the overall model performance, we conducted an ablation study, systematically removing each component and evaluating the resulting impact on segmentation accuracy on the ACDC dataset. Table 5 presents the DSC obtained when individual modules are excluded. The results show that using all proposed modules yields the best performance.

6.2. Memory padding methods

As previously detailed in Section 3.3, during the initial processing stages, the memory bank is either empty or contains insufficient information to perform memory attention effectively, which can compromise the model's performance. Consequently, an ablation study was conducted to evaluate various memory padding strategies and thereby determine the optimal method for these early stages. Five distinct approaches were evaluated on ProstateX dataset:

- **No Padding**: Memory bank remains empty; memory attention is performed based on the number of memory in the memory bank.
- **Padding with Zero**: Initialise the memory bank with zero-value embeddings and follow a FIFO sequence on memory replacement.
- **Padding with One**: Initialise the memory bank with one-value embeddings and follow a FIFO sequence on memory replacement.
- **Padding with Current Image Embedding**: Pad the first few slices of the memory bank with their own image embedding.
- **Padding with NN's Parameters**: Pad the first few slices of the memory bank with an empty neural network's (NN) parameters. Allow the neural network to determine what to pad.

The results presented in Table 6 demonstrate that padding the memory bank with neural network (NN) parameters achieves the highest performance, yielding a Dice Similarity Coefficient (DSC) of 0.7277. This approach surpasses alternatives, with padding using zeros producing a slightly lower DSC, closely following the NN parameter method, while padding with current image embeddings ranks second in performance. Although padding with NN parameters enhances segmentation accuracy, it increases model complexity and computational demands due to the integration of additional trainable parameters. We observed that in Fig. 4, padding with current image embeddings introduces noise in the initial slices, despite its relatively high performance, likely due to inconsistencies in early sequence context. In contrast, padding with zeros provides a simpler yet effective solution, maintaining competitive performance without substantially increasing architectural complexity. Consequently, to strike a balance between efficiency and efficacy, we

Table 3
Dice coefficient and Hausdorff distances (mm) for ACDC; Avg denotes average; RV denotes Right Ventricle; Myo denotes Myocardium; LV denotes Left Ventricle.

Method	DSC				HD95			
	Avg	RV	Myo	LV	Avg	RV	Myo	LV
UNET (2D)	0.852	0.858	0.799	0.899	2.19	3.74	1.41	1.41
UNET (3D)	0.875	0.871	0.806	*0.949*	2.05	3.16	2.00	1.00
TransUNet (2D)	0.867	**0.916**	0.817	0.866	1.55	*1.41*	1.73	1.50
TransUNet (3D)	0.885	0.889	0.8199	**0.948**	1.88	2.24	1.41	2.00
CSA-Net (2.5D)	*0.899*	*0.915*	0.891	0.890	1.47	1.41	2.00	1.00
MedSAM2	0.894	0.902	0.891	0.890	**1.28**	**1.41**	1.41	*1.00*
Ours	**0.907**	0.914	**0.908**	0.898	*1.33*	2.00	**1.00**	**1.00**

Table 4
Dice coefficient and Hausdorff distances (mm) for MSD-Lung.

Methods	DSC	HD95
UNET (2D)	0.795	3.74
UNET (3D)	0.811	3.00
TransUNet (2D)	0.798	2.34
TransUNet (3D)	0.819	3.12
CSA-Net (2.5D)	0.829	*2.27*
MedSAM2	**0.839**	**2.15**
Ours	*0.832*	2.33

Table 5
Importance of model's modules.

Neighbour attention	Memory attention	Mask encoder	DSC
✗	✗	✗	0.730
✓	✗	✗	0.833
✓	✓	✗	0.897
✓	✓	✓	0.907

Table 6
Memory padding methods; Evaluated on ProstateX.

Padding methods	Dice score
No padding	0.6361
Zero	0.7266
One	0.7253
Current image embedding	0.7269
Empty NN's parameters	0.7277

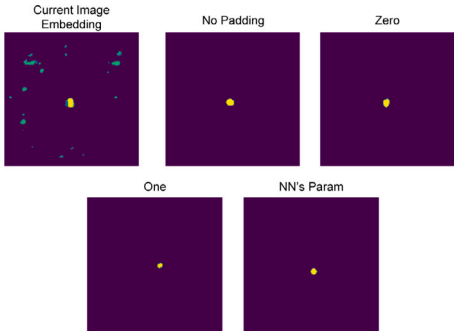

Fig. 4. Results of first slice using different padding methods.

selected padding with zero-value embeddings as the preferred strategy for practical implementation in our 3DAS2D framework.

7. Conclusion

We introduce 3DAS2D, a novel framework that achieves robust 3D medical image segmentation by processing images as a sequence of 2D slices. By integrating inter-slice spatial context using four specialised modules, our method significantly enhances segmentation accuracy, achieve comparable results with state-of-the-art 3D methods in our evaluations. Ablation studies confirm that our proposed components are essential for effectively capturing this volumetric information.

The primary contribution of this work is demonstrating that a 2D-based approach can achieve performance comparable to that of complex 3D models. This finding bridges the gap between 2D and 3D medical imaging, enabling the direct application of powerful, off-the-shelf 2D large vision models to 3D segmentation tasks without requiring any architectural modifications or retraining.

CRediT authorship contribution statement

Yee Zhing Liew: Writing – original draft, Investigation, Formal analysis, Data curation. **Andrew Huey Ping Tan:** Writing – review & editing, Visualization, Supervision, Project administration. **Anwar P.P. Abdul Majeed:** Writing – review & editing, Supervision, Resources, Conceptualization. **Anh Nguyen:** Writing – review & editing, Validation, Supervision. **Paolo Paoletti:** Writing – review & editing, Software, Resources, Project administration. **Wei Chen:** Writing – review & editing, Resources, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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